

Mitigating Spurious Correlations in Natural Language Inference through Ambiguity Filtering and Data Augmentation

Abstract

Natural Language Inference (NLI) models often achieve high performance on standard benchmarks by relying on dataset artifacts rather than true semantic reasoning. In this project, I demonstrate that the SNLI dataset is heavily compromised by these spurious correlations, allowing models to predict correct labels without reading the premise. To address this, I implemented a data-centric pipeline that filters training data based on training dynamics (Dataset Cartography) and augments the remaining subset using back-translation. My experiments show that training on a tiny, curated subset of data (less than 4% of the original) yields a model that is significantly more robust to adversarial attacks than a baseline trained on the full dataset.

1 Introduction

Natural Language Inference (NLI) is a fundamental task in Natural Language Processing, requiring a model to determine whether a hypothesis sentence entails, contradicts, or is neutral with respect to a premise sentence. Pre-trained language models like BERT and ELECTRA have achieved near-human performance on large-scale datasets such as the Stanford NLI (SNLI) corpus.

However, high accuracy on in-domain test sets often masks that models are not necessarily learning to reason. Instead, they frequently rely on surface-level patterns that happen to correlate with labels in the training data but have no causal link to the logical relationship.

For example, crowd-workers tasked with generating "contradictions" often resort to simple heuristics, such as adding the word "not" or identifying a specific entity and saying "nobody" is doing that action. Consequently, a model can learn that the token "nobody" implies contradiction with near 100% probability, ignoring the premise entirely.

In this project, I aim to diagnose and mitigate this reliance on artifacts. The core hypothesis of this work is that *less is more*. By aggressively filtering out the easy examples that reinforce these shortcuts, we can force the model to learn more robust features, even if the total amount of training data is drastically reduced.

The contributions of this report are:

1. **Diagnosis:** Quantifying the extent of artifacts in SNLI using n-gram analysis and a "Hypothesis-Only" baseline.
2. **Filtering:** Applying Dataset Cartography to identify and retain only "Ambiguous" examples where the model struggles.
3. **Augmentation:** Implementing a back-translation pipeline to recover data volume.
4. **Evaluation:** Testing on the Adversarial NLI (ANLI) benchmark to measure true robustness.

2 Related Work

2.1 Hypothesis-Only Baselines

The issue of dataset artifacts in NLI is well-documented. [Poliak et al. \(2018\)](#) conducted a comprehensive study titled "Hypothesis Only Baselines in Natural Language Inference," demonstrating that models can outperform the majority baseline on datasets like SNLI without ever seeing the premise sentence. This finding reveals that the dataset often leaks the label through specific lexical patterns in the hypothesis (e.g., negations or specific entities), effectively reducing the task from logical reasoning between two sentences to a simple text classification task based on the hypothesis alone.

2.2 Data-Centric AI

While much of the deep learning field focuses on improving architectures, my focus is on im-

proving the quality of the data fed into the model. Swayamdipta et al. (2020) introduced *Dataset Cartography*, a method to map training instances based on model training dynamics. They found that training on "ambiguous" instances, those where the model is unsure across epochs, often leads to better generalization than training on easy-to-learn instances. I adopt this framework as the primary mechanism for data selection.

2.3 Adversarial Evaluation

To truly test reasoning, the Adversarial NLI (ANLI) benchmark was created using a human-and-model-in-the-loop process. Annotators were specifically asked to write sentences that fooled a state-of-the-art model. As a result, ANLI penalizes simple keyword heuristics. In this project, I use ANLI Round 1 as the primary metric for robustness.

3 Diagnostic Analysis

Before implementing any mitigation strategies, it was necessary to quantify the severity of the problem in the SNLI dataset.

3.1 Statistical Artifact Discovery

I analyzed the training set to find words that correlate strongly with specific labels. I computed the conditional probability of a label given specific words in the hypothesis sentences.

As shown in Table 1, specific words act as ground-truth labels. For instance, the presence of "nobody" implies a contradiction 99.7% of the time. Similarly, "home sleeping" is a strong indicator of contradiction. This likely occurs because annotators, when given a premise like "A person is outside running," find it easiest to contradict it by writing "A person is home sleeping."

3.2 The Hypothesis-Only Baseline

To confirm that models can exploit these statistics, I trained a model using only the hypothesis sentences. The premises were stripped from the training and test data, forcing the model to rely solely on the hypothesis text. This blind model achieved 60.0% accuracy on the SNLI test set.

Given that SNLI has three balanced classes, a random guess would achieve 33.3%. The fact that a model can achieve nearly double the random performance without even seeing the premise proves that the dataset is fundamentally compromised. The logical relationship has been degraded to a simple keyword matching task.

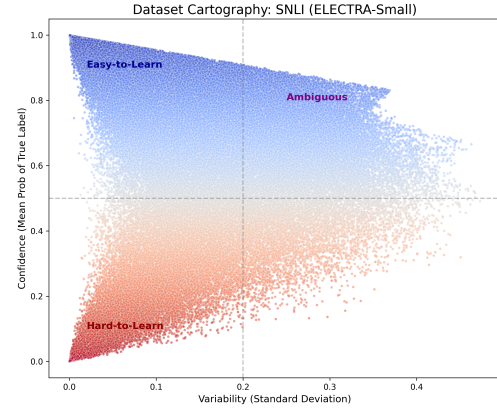


Figure 1: Dataset Cartography Map of SNLI. The dense region in the top left represents easy-to-learn examples (high confidence, low variability). Our mitigation strategy filters for the "Ambiguous" region (high variability) to encourage robust learning.

4 Methodology

Based on the diagnosis, I designed a pipeline to filter out these easy-to-learn artifact examples and augment the ambiguous examples.

4.1 Step 1: Dataset Cartography Filtering

I used the Dataset Cartography framework to map the training dynamics of an ELECTRA-Small model trained on the full SNLI dataset. For every training example, I logged the probability assigned to the true label across all training epochs.

I calculated two metrics for every sentence pair:

- **Confidence:** The average probability the model assigned to the correct label throughout training.
- **Variability:** How much the probability fluctuated across epochs.

I generated a visual map of the dataset (see Figure 1) which revealed distinct regions of data difficulty. According to the Cartography framework, examples with high confidence and low variability are easy-to-learn. These correspond to the artifact examples (e.g., sentences with "nobody"). The model learns them in epoch 1 and never changes its mind because the signal is so obvious.

I applied a filter to remove these easy examples, retaining only the top 33% of examples with the highest variability. These are the ambiguous examples where the model struggled to find a simple correlation, forcing it to actually read the text.

Entailment Word	Count	Prob (%)	Neutral Word	Count	Prob (%)	Contradiction Word	Count	Prob (%)
Some humans	156	98.7	joyously	174	100	Nobody has	338	100
least two	130	96.9	Tall people	108	100	Nobody	2270	99.7
A human	1124	96.4	tall human	587	99.7	Nobody is	1768	99.7
this picture	792	95.8	Tall humans	349	99.4	Two cats	219	99.5
Humans	492	95.3	are siblings	133	99.2	home sleeping	187	99.5
at least	313	94.9	Tall	486	99.2	home watching	116	98.3
than one	252	94.0	are sisters	111	99.1	cats are	216	98.1
picture ,	220	92.7	siblings .	111	99.1	no people	150	98.0
least	390	92.6	A funny	107	99.1	inside watching	149	98.0
wearing clothing	144	92.4	are brothers	138	98.6	No one	429	97.7

Table 1: Top statistical artifacts discovered in the SNLI dataset. Words or bigrams in the hypothesis sentence alone can predict the label with extremely high probability.

4.2 Step 2: Hard Negative Mining

From this ambiguous subset, I further refined the data by creating contrast pairs. I aimed to find pairs of hypotheses for the same premise that look very similar but have different labels.

To do this, I grouped examples by their premise and calculated the Jaccard similarity between hypotheses. The Jaccard similarity measures the overlap between two sets of words (how many words they have in common). I selected pairs where the word overlap was greater than 50%, but the labels were different. This yielded the "Ambiguous Contrast" dataset, consisting of roughly 8,700 examples.

4.3 Step 3: Back-Translation Data Augmentation

The filtering process reduced the training data size by over 98% (from 550,000 to 8,700). Training deep learning models on such small data often leads to instability. To address this, I implemented a back-translation augmentation strategy to artificially increase the size of the curated dataset.

Primary Protocol (Spanish Pivot) I utilized the Helsinki-NLP MarianMT models to perform round-trip translations. My primary pipeline used Spanish as the pivot language. I translated English sentences to Spanish and back to English. Spanish was selected due to its high-resource availability and relatively close syntax to English, which helps preserve the semantic logic required for NLI tasks.

Multilingual Exploration In an attempt to further increase dataset diversity, I also experimented with extending this pipeline to include German and French. The hypothesis was that German’s distinct grammatical structure (e.g., verb placement) might produce more syntactically diverse paraphrases. However, as discussed later in the

Results section, this multilingual expansion did not yield performance improvements over the single-language Spanish baseline. Consequently, the final best-performing model relied solely on the English-Spanish-English augmentation.

Semantic Quality Control (SBERT) Automated translation can introduce noise or alter the meaning entirely (semantic drift). To prevent this, I applied a semantic filter using Sentence-BERT (SBERT). SBERT converts sentences into numerical vectors. I calculated the Cosine Similarity between the original English sentence and the back-translated version.

A generated paraphrase was only added to the training set if it was not identical to the original (to ensure we actually added new data), and the similarity score was greater than 0.85 (to ensure the meaning didn’t change significantly).

This pipeline effectively doubled the size of the curated dataset while maintaining the ambiguity of the examples.

5 Experimental Setup

Hardware and Environment All experiments were conducted on Google Colab using a single T4 GPU. Due to these memory constraints, I used the ELECTRA-Small model. While smaller than BERT-Base, ELECTRA is highly efficient and has shown competitive performance on discriminative tasks.

Hyperparameters I used a batch size of 32 (reduced to 16 for multilingual inference to prevent out-of-memory errors). For the Baseline, I trained for 6 epochs. For the filtered and augmented datasets, I increased the training time to 10 epochs. Since the datasets were so small, the model needed more passes to converge. I used a validation set to monitor loss and saved the best model to avoid

Model Strategy	Data Source	Examples	SNLI (In-Domain)	ANLI (Robustness)
Random Baseline	N/A	N/A	33.3%	33.3%
Hypothesis-Only	Blind	~550k	60.0%	33.8%
Standard Baseline	Full SNLI	~550k	90.1%	32.6%
Full Contrast	Hard Negatives	~84k	84.5%	31.1%
Ambiguous Contrast	Filtered	~8.7k	70.0%	34.5%
Augmented (Spanish)	Filtered + Back-Trans	~16.2k	70.1%	34.9%
Augmented (Multi-Lang)	Filtered + 3 Langs	~31.2k	70.7%	34.0%

Table 2: Main Results. The Standard Baseline collapses on ANLI, performing worse than random guessing. The Augmented (Spanish) model achieves the highest robustness, validating the strategy of filtering plus augmentation.

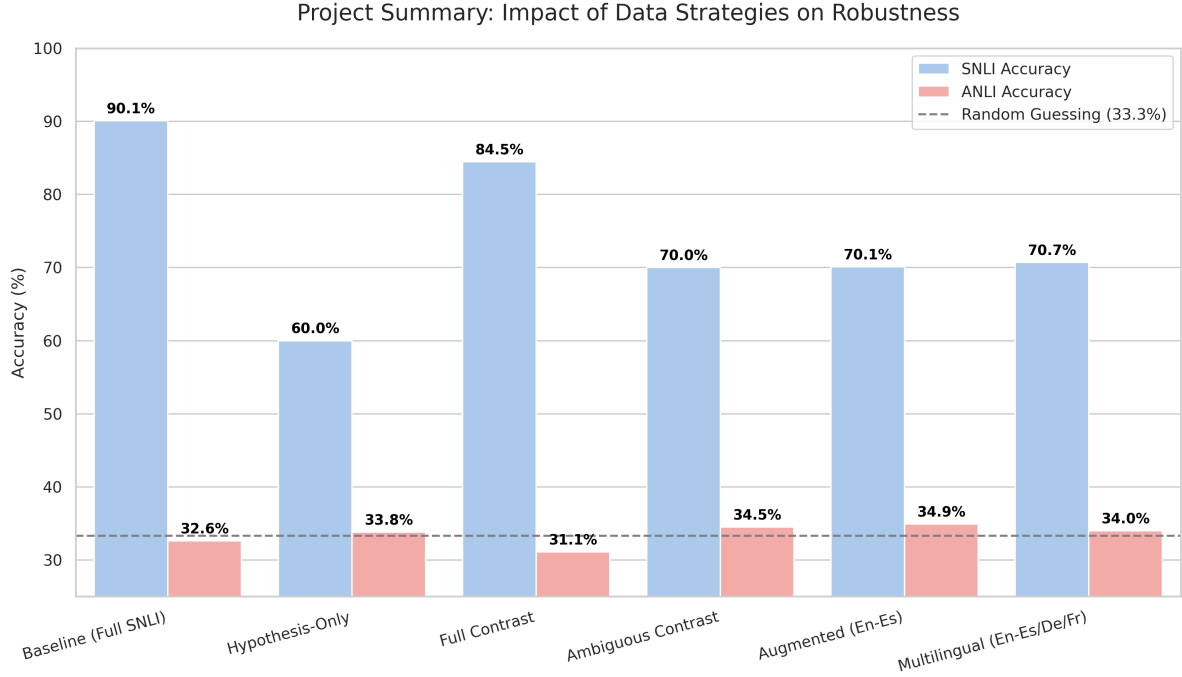


Figure 2: Visual comparison of model performance across baselines and proposed methods. The Augmented Ambiguous model (right) shows the best balance of robustness, exceeding the random baseline on ANLI where standard models failed.

overfitting.

Baselines To validate the effectiveness of my pipeline, I compared against three benchmarks:

1. **Random Baseline:** Since NLI is a three-class task, random guessing yields 33.3% accuracy.
2. **Standard Baseline:** An ELECTRA-Small model trained on the complete, unfiltered SNLI dataset. This represents the standard training paradigm prone to learning spurious correlations.
3. **Unfiltered Lexical Contrast:** A model trained on all examples with high Jaccard similarity (high lexical overlap), but without the Ambiguity Filtering step. This baseline tests

whether simply targeting the "Lexical Overlap" heuristic (McCoy et al., 2019) is sufficient, or if removing easy artifacts via Cartography is also required.

6 Results

6.1 Quantitative Results

Table 2 summarizes the performance across all experiments. The main comparisons are also visualized in Figure 2.

The most striking result is the performance of the Standard Baseline on ANLI (32.6%). Despite achieving 90.1% on the test set it was trained for, it performs worse than random guessing on adversarial data. This empirically proves that the baseline model is overfitting to SNLI artifacts. Instead of

learning logical entailment, the model has latched onto superficial heuristics (such as the presence of negation words) that hold in the training distribution but are explicitly penalized in the adversarial distribution.

My filtered models show a distinct trade-off. The In-Domain (SNLI) accuracy drops significantly (from $\sim 90\%$ to $\sim 70\%$). However, the Robustness (ANLI) score improves to **34.9%**. While a score of 34.9% is low in absolute terms, it represents a meaningful shift. The model has moved from being negatively correlated with the truth (32.6%) to learning robust features (above random chance).

6.2 Qualitative Analysis

To validate the quality of the augmented data, I inspected the generated paraphrases. The primary English-Spanish-English pipeline successfully introduced beneficial lexical diversity without altering the logical labels.

As shown in Table 3, the Spanish back-translation often replaced simple verbs with synonymous phrases (e.g., "playing" $\rightarrow$ "having fun"). This diversity forces the model to map different surface forms to the same semantic concept, preventing it from overfitting to specific tokens like "playing."

Original Hypothesis	Spanish Back-Translation
A person is sitting on a bench.	A person is sitting on a seat .
Two dogs playing in the snow.	Two dogs having fun in the snow.
Nobody is running.	No one is running .

Table 3: Examples from the chosen English-Spanish-English pipeline. The augmentation introduces synonyms while preserving the original logic.

Analysis of Multilingual Failures I also analyzed why the broader multilingual experiment (adding German and French) failed to improve performance. Inspection revealed significant semantic drift, particularly with German.

In one instance, the hypothesis "A man is looking at the ceiling" was back-translated from German as "A man looks at the roof." In NLI, this distinction is critical: if the premise places the man indoors, "ceiling" implies Entailment, while "roof" implies Contradiction. Because the automated SBERT filter only checks for general vec-

tor similarity, it missed these subtle logical flips. This noise explains why the multilingual model performed worse than the Spanish-only model, leading me to exclude German and French from the final proposed pipeline.

7 Discussion

7.1 The Robustness-Accuracy Trade-off

A key finding in the Dataset Cartography framework is the trade-off between in-distribution accuracy and out-of-distribution generalization (Swayamdipta et al., 2020). My results empirically confirm this dynamic. By removing the easy-to-learn data, I removed the shortcuts the model relied upon to achieve high scores on the standard SNLI test set.

The observed drop from 90.1% to 70.1% on SNLI should not be interpreted purely as a performance degradation. As noted by Swayamdipta et al. (2020), high performance on easy-to-learn data often masks a failure to learn the underlying task. The missing 20% likely corresponds to the instances that were previously solved using heuristics rather than reasoning. By forcing the model to unlearn these heuristics, we necessarily lower its performance on the biased test set, but gain the generalization capability required for the adversarial ANLI benchmark.

7.2 Diversity vs. Fidelity in Augmentation

I originally hypothesized that expanding the augmentation pipeline to include German and French would improve robustness by maximizing syntactic diversity. However, the results showed a slight degradation in performance (34.9% $\rightarrow$ 34.0%) compared to the Spanish-only baseline.

This result highlights the trade-off between lexical diversity and semantic Fidelity, a critical tension in data augmentation.

- **Diversity:** German provided structurally distinct paraphrases, which theoretically helps the model generalize better.
- **Fidelity:** However, these distinct structures often introduced semantic drift. As noted in the qualitative analysis (e.g., "ceiling" becoming "roof"), these subtle shifts can invalidate the ground-truth entailment label.

The Spanish-only augmentation represented the optimal sweet spot for this task, providing sufficient lexical variation to prevent overfitting, while

remaining syntactically close enough to English to preserve the precise logical relationships required for NLI.

Nonetheless, even with my best methods, accuracy on ANLI only reached 34.9%. This highlights the extreme difficulty of the ANLI benchmark. ANLI was created by humans specifically attacking RoBERTa-Large models. Expecting an ELECTRA-Small model (which has roughly 10 times fewer parameters) to solve these riddles is ambitious. However, the fact that my data-centric approach beat the baseline (which had access to 50 times more data) validates the methodology.

8 Limitations

There are several limitations to this study inherent to a class project scope:

- **Model Size:** I was restricted to ELECTRA-Small. A larger model might have been able to leverage the noisy multilingual data more effectively.
- **Single-Round Augmentation:** I only performed one round of back-translation. Iterative back-translation might yield better diversity.
- **Reference-Free Evaluation:** I relied on SBERT for filtering. While effective, SBERT itself can have biases. A human-in-the-loop evaluation of a subset of the augmented data would have increased confidence in the dataset quality.

9 Conclusion

This project challenged the prevailing deep learning wisdom that more data is always better. By diagnosing the SNLI dataset, I revealed that a vast majority of its examples act as noise that encourages spurious correlations.

Through a rigorous pipeline of ambiguity filtering and Spanish back-translation, I created a dataset that is 97% smaller than the original but yields a model that is capable of performing above random chance on adversarial data where the baseline fails.

Future work should explore soft filtering (reweighting examples rather than deleting them) and employing LLMs like GPT-4 to generate synthetic hard negatives, which may offer higher quality control than machine translation.

References

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